# Private 5G vs. Wi-Fi in the Enterprise: A Measurement Study of Tunability and QoE

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## Abstract

Enterprise wireless is shifting from best-effort connectivity toward predictable, controllable, and flow-differentiated performance. Wi-Fi remains the dominant enterprise choice, while Private 5G (P5G) is gaining traction as an alternative with a richer set of operator-configurable knobs. Their practical value in real enterprise deployments, however, remains unclear. In this paper, we present a comparative measurement study of P5G and Wi-Fi on a commercial enterprise testbed, evaluating single-user performance, multi-user scaling, flow prioritization, and application QoE. Our results show the two technologies offer complementary strengths. Wi-Fi leads in single-user scenarios, but P5G's TDD configurability can match or exceed it in specific traffic directions. Under multi-user load, P5G's centralized scheduler delivers better fairness and stable latency. P5G's flow prioritization protects high-priority flows without starving background traffic, unlike Wi-Fi's winner-take-all flow priority model.

## CCS Concepts

• **Networks** → **Mobile networks**; **Network measurement**; **Network performance analysis**.

## Keywords

Private 5G; Wi-Fi; Wireless Networks; Measurement; Network Performance;

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## 1 Introduction

Emerging enterprise workloads, such as XR, video conferencing, industrial automation, real-time monitoring, require high bandwidth, low latency, and traffic differentiation. These requirements push enterprise wireless beyond simple capacity maximization toward controllability.

Wi-Fi remains the dominant enterprise wireless technology, but its contention-based access model has fundamental limits: flows compete independently with no per-flow isolation, making performance unpredictable under load. Private 5G (P5G) is gaining traction as an enterprise alternative [4]. P5G uses a centralized scheduler and exposes operator-configurable knobs designed for predictable, controllable, and flow-differentiated performance. Their practical value in real deployments, however, remains unclear.

Among the many operator-configurable knobs P5G exposes, we focus on three of them. The **TDD slot ratio** (Time Division Duplex) controls how the air interface divides each frame between downlink and uplink slots. The **5QI (5G QoS Identifier)** is a flow-level class that sets scheduling priority and packet delay budget. Finally, **AMBR (Aggregate Maximum Bit Rate)** caps the throughput of a prioritized flow such that background traffic retains access to residual capacity.

These mechanisms are implementation-dependent: their effectiveness is shaped by scheduler internals, core-network policies, and commercial hardware behavior, none of which can be inferred from standards alone. Prior work on P5G has largely focused on testbed architecture [2, 3, 12, 14] or theoretical slicing [11]; comparisons with Wi-Fi have mostly used public MNO networks [9]. Ghosh et al. [5] compare P5G and Wi-Fi in an enterprise setting but only provide a high-level overview without characterizing individual knobs. Prior work on dynamic TDD adaptation [7, 8] demonstrates the efficacy of the TDD slot ratio knob in shifting DL/UL bandwidth allocation, but is limited to SDR testbeds or simulation without comparison against enterprise Wi-Fi.

In this paper, we ask: *how effective are P5G's operator-configurable knobs compared to Wi-Fi's QoS mechanisms, and what tradeoffs do they introduce for performance, fairness,*

*and application QoE?* We evaluate both technologies across single-user performance, multi-user scaling, explicit flow prioritization, and application-level QoE.

Our results show that P5G and Wi-Fi offer different strengths.

• **Single-user performance.** Wi-Fi outperforms P5G in both throughput and latency under no background load. P5G, however, offers a configurable TDD mechanism that enables flexible downlink (DL) and uplink (UL) resource allocation. A DL-heavy TDD configuration can match or even exceed Wi-Fi's downlink throughput, while a UL-heavy configuration reduces the gap in uplink throughput. In contrast, Wi-Fi lacks an equivalent mechanism for trading off DL and UL performance.

• **Multi-user scaling.** P5G's scheduler ensures more predictable per-device fairness compared to Wi-Fi. While Wi-Fi RTT inflates by up to 6× with competing flows, P5G maintains stable latency. This behavior is an inherent consequence of scheduled medium access rather than a configurable capability.

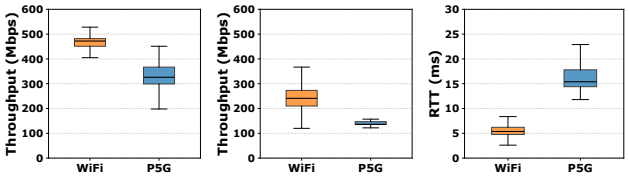
• **Flow prioritization.** Wi-Fi's QoS mechanism effectively isolates a priority flow but provides no protection for background flows, which can be completely starved as the priority flow retains channel access regardless of its actual utilization. In contrast, P5G's 5QI-based prioritization and AMBR rate limiting achieve comparable traffic isolation while maintaining fairness for background flows. A DASH video case study further demonstrates that these differences translate into measurable improvements in QoE.

# 2 Experimental Methodology

## 2.1 Testbed and Devices

Our testbed consists of two parallel wireless deployments in the same indoor lab environment: an industry-grade HPE Aruba Wi-Fi 7 access point (AP) and a Sercomm MosoLabs private 5G (P5G) base station (BS). Both the Wi-Fi AP and the P5G BS were connected over Ethernet to a dedicated laptop that hosted the application servers. For the P5G deployment, the same laptop also ran the Open5GS core. We used six rooted Google Pixel smartphones as user equipment (UE): four Pixel 10 Pro devices, one Pixel 9, and one Pixel 8 Pro. All phones were placed approximately 6 ft from the AP/BS in a controlled indoor lab environment.

The Wi-Fi 7 access point operates in the 5 GHz band with a 40 MHz channel bandwidth, while the P5G base station is configured on the n48 CBRS band at 3.7 GHz, also with 40 MHz bandwidth. The n48 band operates in TDD mode, which is what enables the configurable DL/UL slot ratio knob we study in §3.2. Both deployments therefore used the same channel bandwidth, allowing us to compare their behavior under similar spectrum allocations. Both the Wi-Fi access point and the P5G base station support 4×4 MIMO, while the smartphones use 4×4 MIMO for 5G-sub 6 transmissions, and 2×2 MIMO for Wi-Fi.




**(a) Downlink.** **(b) Uplink.** **(c) RTT.**

**Figure 1: P5G vs. Wi-Fi: Baseline performance.**

## 2.2 Test Suite

**Throughput and RTT Measurements.** We used `iperf3` to generate TCP traffic in both downlink and uplink directions. We logged per-flow throughput at 100 ms intervals. For latency measurements, we use ICMP-based PING with a 10 ms ICMP Echo request interval. For each experiment, we collected 3–5 traces of 20–200 s each and present boxplots over all logged samples across the repeated runs.

**Video Streaming Evaluation.** We evaluated video streaming performance using a custom web-based DASH player built on dash.js, a widely used JavaScript implementation of MPEG-DASH playback. The player was configured with the default rate-based adaptive bitrate (ABR) algorithm.

We used two video sequences encoded in DASH video-on-demand format using FFmpeg, with a segment duration of 4 s. Both sequences used the same 11-level bitrate ladder, ranging from 3 Mbps to 80 Mbps, and covering resolutions from 360p to 2160p at 30 and 60 fps. We use the same encoding ladder across the Wi-Fi and P5G deployments to ensure that ABR decisions are comparable across technologies. Each streaming session lasts 2 minutes, and the player buffer is capped at 40 s. We report stall count, stall duration, bitrate distribution, and resolution distribution as application-level QoE metrics.

# 3 Single-User Performance

We begin by characterizing the single-user performance of Wi-Fi and P5G. This provides a best-case baseline for both technologies before introducing contention and application-level workloads. We then investigate how P5G's first tunable knob, the TDD configuration, affects this baseline. All experiments in this section use a single active device.

## 3.1 Baseline performance: Wi-Fi and P5G

Fig. 1 shows the single-user baseline performance across downlink throughput, uplink throughput, and RTT.

**Downlink.** Wi-Fi achieves substantially higher downlink throughput than P5G. Wi-Fi achieves downlink throughput, P25/P50/P75, of 451/472/482 Mbps compared to

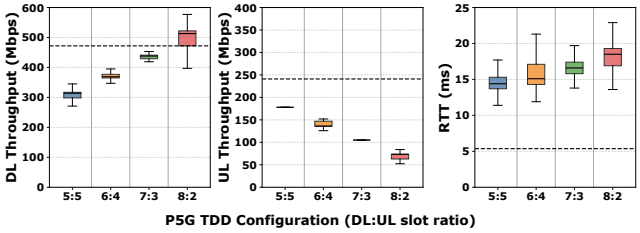



**Figure 2: Impact of different P5G TDD configurations on performance.**

299/326/367 Mbps for P5G. P5G also shows higher variability, with an interquartile range (IQR) of 68 Mbps compared to 31 Mbps for Wi-Fi. Thus, under single-user conditions, Wi-Fi provides both higher median downlink throughput and more stable downlink performance.

**Uplink.** Wi-Fi also outperforms P5G in uplink throughput. Specifically, Wi-Fi achieves P25/P50/P75 throughputs of 210/241/273 Mbps compared to 136/136/147 Mbps for P5G. Both technologies deliver lower uplink than downlink throughput — an asymmetry likely attributable to the smartphone's lower transmit power and smaller form-factor antennas relative to the AP/BS, and commonly observed in prior measurement studies [1, 6, 10]. For P5G, the default TDD configuration (6:4 DL:UL) further amplifies this asymmetry by allocating fewer slots to the uplink direction. We study the impact of this knob in §3.2. Notably, P5G uplink is considerably more stable than Wi-Fi: its IQR spans only 11 Mbps, while Wi-Fi's uplink IQR is 63 Mbps.

**RTT.** Wi-Fi achieves substantially lower RTT than P5G, with P25/P50/P75 of 5–6 ms for Wi-Fi vs. 14–18 ms for P5G, an almost 3× difference. This discrepancy stems from P5G's request-grant scheduling overhead [5]: a device must issue a scheduling request and wait for an uplink grant before transmitting, adding several milliseconds even when the server is co-located. Wi-Fi's contention-based access, by contrast, gives a single uncontested device near-immediate channel access.

## 3.2 TDD Configuration

In the previous section, we observed that Wi-Fi outperforms P5G across all metrics in single-user conditions. We now ask whether P5G's TDD configuration knob reduces that gap. P5G uses Time Division Duplex (TDD) scheduling, where the ratio of downlink to uplink slots within each frame is configurable. We were able to set the following downlink (DL)-to-uplink (UL) slot ratio TDD configurations: 5:5, 6:4 (default), 7:3, 8:2. Fig. 2 compares the resulting downlink throughput, uplink throughput, and RTT across these TDD configurations against the median Wi-Fi baseline (shown as dashed lines).

**Downlink.** DL throughput increases monotonically (Fig. 2 (left)) as the configuration becomes more DL-heavy: from ∼310 Mbps at 5:5 to ∼510 Mbps at 8:2 in the median case. More importantly, the 8:2 configuration exceeds the Wi-Fi median (∼472 Mbps), and 7:3 comes close at ∼438 Mbps. This shows that TDD configuration is an effective knob for closing the P5G DL gap with Wi-Fi.

**Uplink.** The UL throughput trend (Fig. 2 (middle)) is the inverse: median throughput drops sharply from ∼180 Mbps at 5:5 to ∼70 Mbps at 8:2. No P5G configuration reaches the Wi-Fi UL median (∼241 Mbps). The 5:5 split represents the practical UL ceiling: in 5G NR TDD, the base station delivers scheduling grants to devices via downlink slots, so a minimum number of DL slots must be preserved for the grant-transmission exchange to function correctly. This means the configuration cannot be pushed further in the UL direction, leaving P5G UL throughput structurally below its Wi-Fi counterpart.

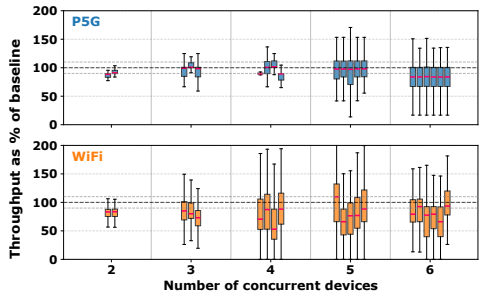
**RTT.** As the number of downlink slots increases, RTT degrades monotonically (Fig. 2 (right)): the growing scarcity of uplink slots forces devices to wait longer before an uplink transmission opportunity becomes available. Consequently, the 5:5 configuration, being the most balanced, achieves the lowest median RTT of ∼14 ms, while 8:2 reaches ∼19 ms. Regardless of configuration, P5G RTT remains 3× above the Wi-Fi median of ∼5 ms. The request-grant scheduling overhead is inherent to cellular systems and cannot be eliminated even by tuning the TDD pattern.

# 4 Multi-User Scaling

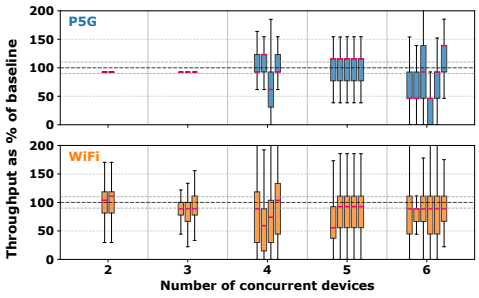
## 4.1 Throughput

The single-user results in §3 establish best-case performance for each technology. We now examine how both networks behave under concurrent load, scaling the number of simultaneously active devices from two to six, each running a backlogged TCP flow. Rather than reporting absolute throughput, we normalize each device's throughput to the ideal fair share, defined as the single-user capacity divided by the number of active devices. Fig. 3 shows this ratio as a function of the number of concurrent devices.

**Downlink (Fig. 3a).** P5G's individual median throughput remains within ∼10% of the ideal fair share across 2–5 concurrent devices. At 6 devices, the lowest median drops to ∼80% of fair share, suggesting scheduler overhead at maximum device count. Variance is low at 2–3 devices and grows gradually at 5–6, reflecting the increased scheduling complexity. In contrast, for Wi-Fi, even at 2 devices, median throughput already falls to ∼83% of fair share, dropping further to 50–75% range for 4–5 devices, with substantially wider IQRs throughout. This is expected from a contention-based random access mechanism where flows compete for the channel

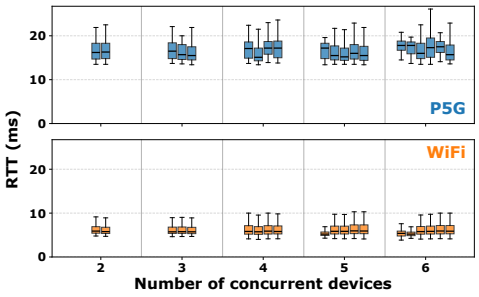



(a) Downlink.

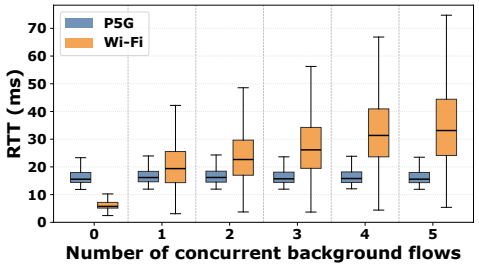



(b) Uplink.

**Figure 3: Per-device throughput as a fraction of the single-user baseline as concurrent users scale.**




(a) RTT as concurrent users scale.




(b) RTT under competing backlogged flows.

**Figure 4: RTT under multi-user load.**

independently, leading to higher variance and less consistent per-device fairness.

**Uplink (Fig. 3b).** For P5G, the scheduler performs well when there are few users actively transmitting: at 2-3 devices, individual flows consistently achieve ~90% of fair share with very tight distributions. At 4 or more devices, both fairness and stability degrade noticeably, some flows receive over 120% while others drop to ~60%, suggesting the uplink scheduler is less effective under higher load than the downlink. Wi-Fi uplink shows high variance with wide IQRs even at 2 devices and is more unpredictable than P5G at lower device counts; however, at 5–6 devices both technologies exhibit comparable variability.

## 4.2 Latency

Similar to the throughput experiments, we measure the per-device RTT as the number of concurrently active devices scales from two to six. Fig. 4a shows the RTT distribution per device.

Both technologies maintain stable RTT throughout: P5G stays at ~15–17 ms and Wi-Fi at ~5–6 ms across all device counts, preserving the same ~3× gap observed in the single-user baseline. For P5G, this is expected; the centralized scheduler allocates resources independently of other devices' activity, so a PING flow is unaffected by co-existing flows. For Wi-Fi, PING packets are small (38 Bytes) and triggered every 10 ms, so collision probability among competing PING flows remains negligible – thereby not inflating the RTT.

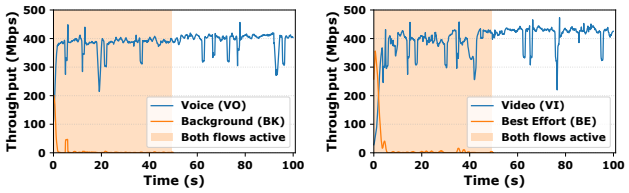
However, we notice a significant difference in behavior when PING flows compete with *backlogged* TCP traffic rather than other PING flows. Fig. 4b shows the RTT of a single PING flow as a function of increasing number of competing backlogged TCP flows.

As the number of background flows increases, P5G RTT remains unchanged. The P25/P50/P75 RTT is around 14/16/18 ms across all conditions ($n = 0$ through $n = 5$). Similar to Fig. 4a, P5G's scheduler allocates resources per flow independently, so heavy TCP traffic competing for bandwidth cannot displace a PING packet from its allocated transmission opportunity.

Wi-Fi behaves very differently. With no background traffic, the median RTT is ~6 ms. Adding just one backlogged flow causes the median to jump to 19.4 ms (P25/P75: 14.3/25.5 ms). The RTT continues to inflate as the number of background flows increases, reaching 33.1 ms (P25/P75: 24.1/44.4 ms) with five background flows, a ~6× increase over baseline. In Wi-Fi, all flows share the same contention-based channel and as heavy TCP flows keep the medium continuously busy, the small PING packet loses channel access repeatedly, inflating the RTT.

# 5 Flow Prioritization

The previous sections demonstrated that Wi-Fi outperforms P5G under single-user conditions. However, under multi-user scenarios, both technologies exhibit performance instability, high variance, and fairness issues. Enterprise settings increasingly demand customized performance for different




(a) VO vs. BK. (b) VI vs. BE.

**Figure 5: Wi-Fi flows with different AC contending.**

flows or user groups. A key advertised advantage of P5G is its ability to isolate and prioritize traffic through different network slicing mechanisms. Wi-Fi, too, offers different QoS mechanisms. In this section, we evaluate how effectively both technologies support flow prioritization in practice.

## 5.1 Wi-Fi flow prioritization

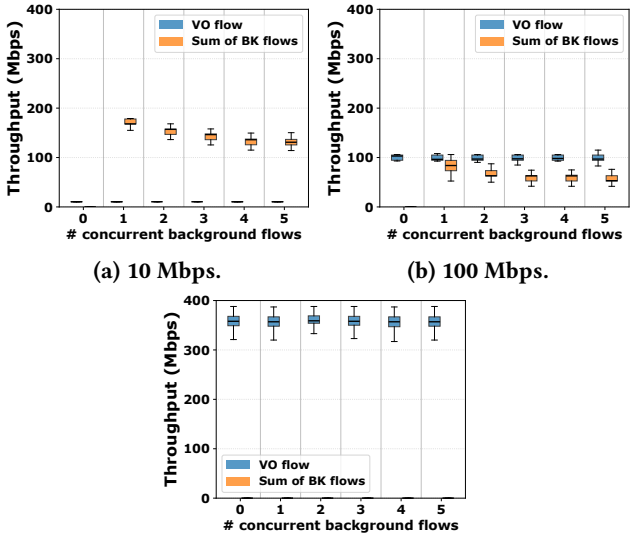
Wi-Fi provides flow prioritization through *Wi-Fi Multimedia* (WMM), which defines four Access Categories (ACs) in decreasing priority order: Voice (VO) > Video (VI) > Best Effort (BE) > Background (BK). Each AC uses distinct EDCA (Enhanced Distributed Channel Access) parameters (a smaller contention window and shorter inter-frame spacing for higher-priority categories) giving the higher priority AC more frequent channel access than the lower priority ones.

To understand how WMM priority plays out under contention, we ran two backlogged iperf3 flows simultaneously: one active for 100 s (the persistent flow) and a second active for only 50 s (the competing flow), each marked with a different WMM AC. Fig. 5a and Fig. 5b show two representative examples. In both cases, the higher-priority flow occupies the full link capacity while the lower-priority flow is almost entirely starved during the co-existence window.

The above experiment establishes the priority effect with a single background flow, both running backlogged traffic. We now extend this in two dimensions: (i) scaling the number of background BK (lowest priority) flows from 1 to 5, and (ii) varying the application-level sending rate of the isolated VO (highest priority) flow (10 Mbps, 100 Mbps, backlogged) to examine what happens when the high-priority flow demands less than the full link capacity. Notably, Wi-Fi WMM provides no mechanism to cap the rate of the prioritized flow, so the VO flow holds its channel priority regardless of how much bandwidth it actually consumes. Fig. 6 shows the results.

Across all configured rates, the isolated VO flow consistently achieves its application sending rate with low IQR (≤10–20 Mbps) regardless of the number of background flows. The prioritization is stable and rate-agnostic.

However, this prioritization comes at a significant cost to the background BK flows. Even at low VO rates, the background flows are heavily penalized. At 10 Mbps VO flow, adding a single BK background flow results in that flow receiving only ~170 Mbps — meaning the combined usage of




(a) 10 Mbps. (b) 100 Mbps.

(c) Backlogged.

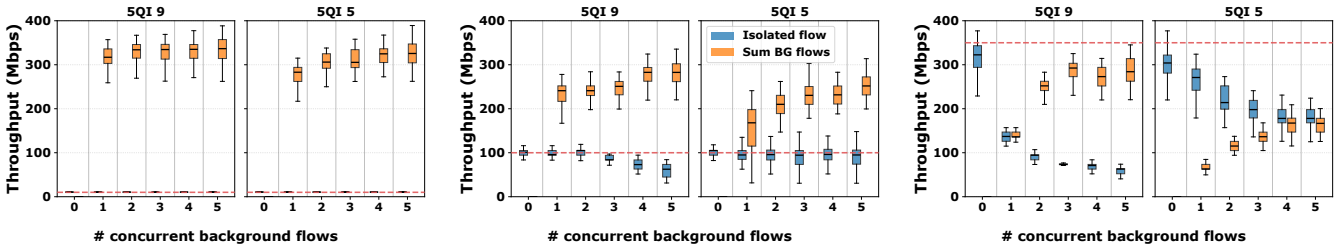
**Figure 6: Wi-Fi prioritized flow throughput under varying number of concurrent background flows for different source rates.**

both flows is ~180 Mbps, while the VO flow alone can achieve ~400 Mbps (Fig. 5a). The VO flow's EDCA priority effectively blocks the BK flow from accessing the remaining ~220 Mbps, leaving it idle. This is a fundamental inefficiency of WMM: even when the high-priority flow is rate-limited at the application layer, the channel-level priority reservation prevents the lower-priority flow from reclaiming unused capacity.

Furthermore, the aggregate background throughput decreases as more BK flows are added. BK flows operate with a larger contention window than VO, hence, as more BK flows compete for channel access, collision probability among them increases, causing each to back off longer and receive a smaller share of the already-limited residual bandwidth. At 100 Mbps, the aggregate BK throughput drops from 84 Mbps at $n = 1$ to 53 Mbps at $n = 5$. At the backlogged rate, the BK flows are completely starved across all background counts.

## 5.2 P5G flow prioritization

In P5G, one way to separate and prioritize flows is to assign different *5G QoS Identifier* (5QI) values. 5QI is a scalar index defined in 3GPP [13] that maps a flow to a set of QoS characteristics, including scheduling priority and packet delay budget. By default, Open5GS assigns flows to 5QI 9, a standardized QoS class characterized by low scheduling priority and a packet delay budget of 300 ms, making it suitable for general-purpose traffic such as web browsing and file transfers. Assigning a higher-priority class, e.g., 5QI 5, elevates the scheduling priority and tightens the packet delay budget to 100 ms, causing the network to treat that flow more favorably.




(a) 10 Mbps. (b) 100 Mbps. (c) Backlogged (5QI 5 350 Mbps AMBR).

**Figure 7: Isolated flow (blue) and sum background flows (orange) throughput under 5QI 9 vs. 5QI 5 (with AMBR) for different rates, as the number of concurrent background flows increases.**

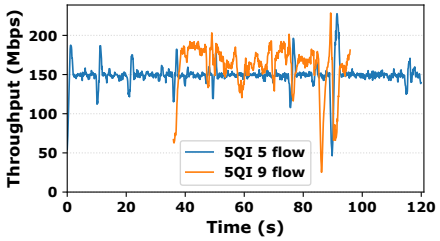



**Figure 8: P5G timeline with a 5QI 5 flow (AMBR 250 Mbps, application rate 150 Mbps) and a backlogged 5QI 9 flow introduced at $t \approx 36$ s.**

Unlike Wi-Fi, where a high-priority flow can consume the full link capacity regardless of background traffic demand, P5G enables the operator to cap the priority flow's throughput using an Aggregate Maximum Bit Rate (AMBR) policy, thereby preserving bandwidth for background traffic and preventing starvation. We observed that the Open5GS AMBR shaper is imperfect: enforcement is tight at low rates (≤10 Mbps), but above 50 Mbps the achieved throughput runs ~10–16% below the configured ceiling (e.g., a 100 Mbps AMBR yields ~84 Mbps median), suggesting shaper overhead in the core network.

Next, we examine how the 5QI 5 flow behaves under competing background traffic (backlogged 5QI 9 flows) for different configured AMBR rates. For comparison, we repeat the experiment with an unprioritized 5QI 9 flow competing against backlogged 5QI 9 background flows. In this case, no AMBR is applied to the unprioritized flow; instead, its transmission rate is controlled directly by the iperf3 application. Fig. 7 shows, for multiple configured rates, the throughput of the prioritized 5QI 5 flow and the unprioritized 5QI 9 flow as the number of concurrent background flows scales from 0 to 5. We also show the aggregate throughput of the background 5QI 9 flows in each case.

**Low rate (10 Mbps).** There is practically no difference between the two flow types at this rate. Both achieve the full configured rate (Fig. 7a) regardless of the number of background flows. This is expected: P5G's single-user capacity is ~360 Mbps, hence even with 5 background flows sharing the remaining capacity, each receives ~70 Mbps, leaving no shortage for a 10 Mbps isolated flow.

**Medium rate (100 Mbps).** As the configured rate increases to 100 Mbps, a clear difference emerges (Fig. 7b). The median 5QI 9 throughput starts to degrade as background flows are added: from 96 Mbps at $n = 1$ down to 62 Mbps at $n = 5$, as the scheduler distributes capacity equally among all contending flows. The 5QI 5 flow, by contrast, holds steady at ~96 Mbps in the median case across all background counts, demonstrating effective isolation by the scheduler.

**High rate (max rate).** The gap widens substantially at higher rates. At the maximum offered rate (Fig. 7c), the median throughput of the 5QI 9 flow decreases from 137 Mbps at $n = 1$ to 63 Mbps at $n = 5$, whereas the 5QI 5 flow drops only from 271 Mbps to 178 Mbps, remaining approximately 3× higher under maximum load.

Notably, the 5QI 5 isolation does not fully suppress the background flows. Background flows collectively achieve ~166 Mbps under 5QI 5 vs. ~283 Mbps under 5QI 9, reflecting the scheduler's fair treatment of background traffic within the remaining capacity after honoring the 5QI 5 allocation. In addition, P5G avoids wasting capacity when the prioritized flow's application rate is below its AMBR ceiling. Fig. 8 shows a two-flow experiment where the 5QI 5 flow is configured with a 250 Mbps AMBR but sends at only 150 Mbps. When a backlogged 5QI 9 flow joins at $t \approx 36$ s, the scheduler immediately allocates the unused ~100 Mbps to it. In contrast, Wi-Fi's EDCA priority reserves channel access regardless of how much the priority flow actually uses, with no mechanism to reclaim idle capacity for lower-priority flows.

One caveat, however, is that the 5QI 5 traffic isolation comes with noticeably higher throughput variability, especially at higher rates. At max rate (Fig. 7c), 5QI 5 IQR spans 30–53 Mbps versus 2–21 Mbps for 5QI 9, likely a consequence of the imperfect AMBR shaper at the core network.

## 6 Impact of flow prioritization on applications

We examine how the flow prioritization mechanisms studied in §5 translate to measurable improvements in application-level Quality of Experience (QoE). We measure video streaming performance for P5G and Wi-Fi under three conditions:

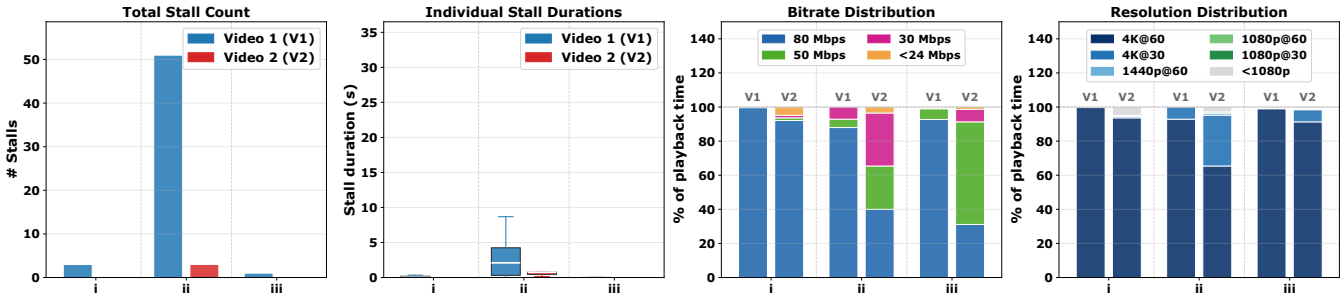



**Figure 9: P5G QoE for videos V1 and V2 under three conditions: (i) solo video, (ii) video with competing backlogged flows (no priority), and (iii) video with competing flows marked as 5QI 5 and AMBR 150 Mbps.**

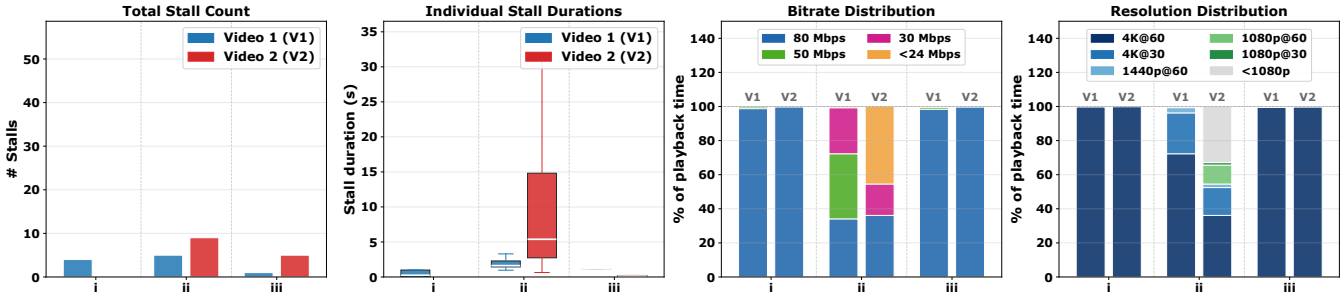



**Figure 10: Wi-Fi QoE for videos V1 and V2 under three conditions: (i) solo video, (ii) video with competing backlogged flows (no priority), and (iii) video with competing flows marked as Voice (VO) AC.**

(i) a single video flow with no competing traffic, (ii) a video flow competing with heavy backlogged background traffic and no priority marking, and (iii) the same setup as (ii) but with the video flow marked as high-priority.

For Wi-Fi, we assign the video flow to the Voice (VO) Access Category and background flows to Background (BK). As shown in §5, a higher-priority AC completely suppresses lower-priority ones, so VO gives the video flow maximum channel access protection while BK places the background flows in the lowest contention tier. For P5G, we configure the video flow as 5QI 5 with an AMBR of 150 Mbps, well above the maximum video bitrate of 80 Mbps to ensure the video flow is never rate-limited by the ceiling, while still bounding its allocation to protect background flows from being fully starved. Two different videos (V1 and V2) are streamed and we report four QoE metrics: total stall count, individual stall duration, bitrate distribution, and streaming resolution.

**P5G.** Fig. 9 shows the video streaming QoE metrics for the three conditions for P5G. In the solo condition (i), both videos achieve near-ideal QoE: V1 is streamed at 80 Mbps for ~99.6% of the time at 4K@60 resolution with only 3 stalls, each lasting under 0.3 s; V2 experiences zero stalls with 92% of playback at 80 Mbps. This is consistent with the single-user P5G capacity of ~360 Mbps (§3) being sufficient for 80 Mbps bitrate videos.

When background flows are introduced without priority marking in condition (ii), we observe severe degradation in QoE metrics for both videos. For V1, the player selected the maximum bitrate (80 Mbps) 87.9% of the time instead of adapting to a lower bitrate. However, this resulted in 51 stall events (median stall duration of 2.1 s, reaching up to 8.7 s), because the network intermittently failed to sustain the required throughput in the presence of competing background flows. For V2, the player, by contrast, adapted aggressively: it experiences only 3 brief stalls but trades quality instead, with 60% of playback falling below 80 Mbps (~35% at 30 Mbps or below) and the streaming resolution dropping from 4K@60 to 4K@30 for nearly 30% of the session.

In condition (iii), assigning 5QI 5 with a 150 Mbps AMBR to the video flows dramatically improves QoE for both videos. V1's stall count drops from 51 to 1, and its bitrate holds at 92.7% at 80 Mbps with 4K@60 resolution. V2 eliminates stalls entirely, and the fraction of time spent at 4K@60 recovers to 91.3%. However, its bitrate does not fully return to the solo-case level, with the player selecting a bitrate below the maximum more than 70% of the time. The priority marking shields the flow from stalls and resolution drops, but

unlike the solo condition, background flows are still present and competing. As shown in Fig. 7, 5QI 5 flows under contention exhibit significant throughput variability (IQR of 30–82 Mbps). In response to this instability, the ABR player conservatively selects a lower bitrate tier, operating primarily at 50 Mbps rather than risking playback stalls at 80 Mbps. Nevertheless, P5G's prioritization delivers clear and measurable QoE protection relative to the unprotected case.

**Wi-Fi.** Fig. 10 shows the Wi-Fi results. In the solo condition (i), both videos achieve near-ideal QoE at 80 Mbps and 4K@60: V1 has 4 brief stalls (median 0.2 s) and V2 experiences zero stalls.

Under competing background traffic in condition (ii), both videos degrade substantially. V1's bitrate collapses: only 34% of playback is at 80 Mbps, with 38% at 50 Mbps and 27% at 30 Mbps, and resolution drops to 4K@30 for 24% of the time. V2 suffers more severely in terms of stall intensity: 9 stalls with a median duration of 5.4 s reaching up to ∼30 s, while 45% of playback falls to sub-24 Mbps and resolution drops below 1080p for 33% of the session.

Applying WMM Voice (VO) priority in condition (iii) produces a near-complete recovery for both videos. V1's bitrate is 80 Mbps 98.4% of the time and 4K@60 resolution with just 1 stall. V2 recovers fully to 99.6% at 80 Mbps and 4K@60; its 5 residual stall events last a median of only 0.1 s each. Notably, unlike the P5G case where throughput variance under contention caused the ABR player to conservatively settle at 50 Mbps, Wi-Fi's EDCA priority grants the VO flow unrestricted channel access, allowing the player to consistently select the highest bitrate tier. As established in §5, however, this recovery comes at the complete cost of starving the background BK flows.

## 7 Conclusion

We presented a comparative measurement study of P5G and Wi-Fi on a commercial enterprise testbed, evaluating the effectiveness of P5G's operator knobs, including TDD slot ratio, 5QI, and AMBR, against Wi-Fi's QoS mechanisms. Wi-Fi leads in single-user throughput and latency, but P5G's TDD knob can shift the downlink/uplink balance to match or even exceed Wi-Fi in one direction, a tradeoff for which Wi-Fi has no equivalent control knob. Under multi-user load, Wi-Fi RTT inflates by up to 6× while P5G remains stable, and P5G's scheduler delivers more fair per-device throughput, particularly at lower device counts (2–4 DL, 2–3 UL). Wi-Fi's WMM starves background flows even when the priority flow is rate-limited; P5G's 5QI/AMBR provides comparable isolation while preserving residual capacity for background traffic. The impact of flow prioritization is further evident in application QoE: P5G's prioritization reduces stalls by 50× and sustains full bitrate for over 90% of the session, whereas Wi-Fi's prioritization restores QoE to that of a solo flow with no background traffic, but at the complete expense of background flows.

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